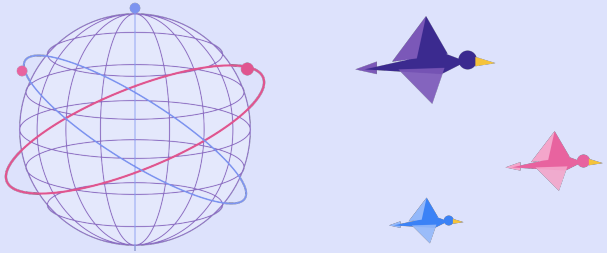
PARTICIPANT PRESENTATION

Qiskit Fall Fest 2026

Hackathon · Team Details



Participant Details

PROJECT TITLE
Q-matrix · Quantum Space

TRACK
Quantum Education & Learning

TEAM NAME
Q-matrix

TEAM LEADER
Chinmay R

EMAIL / CONTACT
github.com/ChinmayRERC

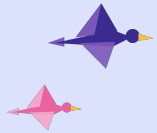
TEAM MEMBERS
Chinmay R · M Naveen · Thulasi Ram M

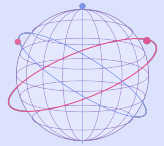
INSTITUTION / ORGANISATION
Visvesvaraya Technological University

COUNTRY
India

TEAM CODE VTU-QFF-HC-0024

Problem Statement





THE GAP

- Quantum courses explain amplitudes, but learners cannot see them; notebooks and cloud queues add friction before the first insight.
- Generic AI tutors hallucinate circuits and cannot read the learner's live state or mistake.
- Instructors receive completion ticks, not evidence of conceptual change.

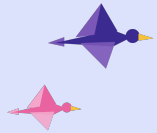
WHO FEELS IT

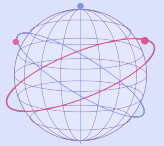
- Students in low-resource colleges without lab hardware or paid cloud credits.
- Self-learners who stall between a video and a working Qiskit program.
- Teachers who need a defensible mastery signal, not another quiz score.

DESIGN GOAL

- One browser workspace: learn → build → simulate → export real Qiskit.
- Grounded feedback from the circuit itself; every secret stays server-side.

Objectives & Novelty





OBJECTIVES

- Make quantum literacy measurable: a 7-stage loop from intuition to Qiskit export.
- Ground every tutor answer in the live circuit, simulator state and static analysis.
- Give educators an auditable mastery trail: BKT posterior, gain and challenge record.

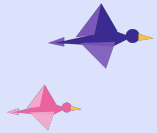
NOVELTY

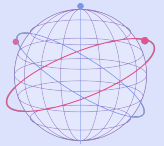
- Agent-reviewed content pipeline: deterministic analyzer + NVIDIA Nemotron tutor.
- Challenge mode loads scaffolds, never solutions; Reset restores a clean start.
- Browser-native density simulation with noise, mid-circuit measurement and feed-forward.

SCALE

- 176 topics · 20 domains · 8 learning paths · 13 circuit templates · 11 challenges.
- 147 automated checks, including 15 Qiskit 2.5.2 reference-algorithm tests.

Solution & Quantum Advantage





WHAT WE BUILT

- Q-matrix is the app, Quantum Space is the platform, Chinnu is the grounded tutor.
- Learn → Compose → Practice → Galaxy → Progress: one continuous learning loop.
- React SPA + FastAPI on Render; GitHub Pages demo; NVIDIA Nemotron 3.5 Lightning server-side.

QUANTUM ADVANTAGE

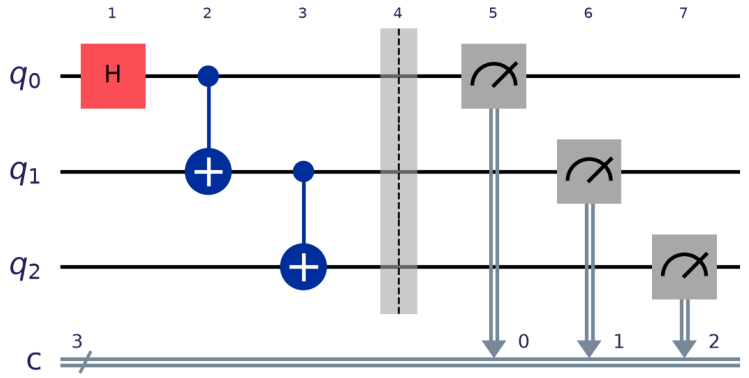
- Real statevector/density simulation exposes superposition, interference, entanglement and measurement.
- Noise and feed-forward are first-class: teleportation and mid-circuit correction run in-browser.
- Every artifact is portable: OpenQASM 2.0, Qiskit, Cirq and PennyLane exports.

WHY NOW

- Qiskit 2.5.2 maturity + browser compute + grounded LLMs = a real quantum classroom.

Algorithm & Circuit Design

GHZ preparation · three qubits, two CNOTs



H creates superposition; CNOTs spread one shared phase across q0–q2. Measurement outcomes are perfectly correlated.

ALGORITHM

- GHZ: H(q0), CNOT(q0,q1), CNOT(q1,q2) → (|000⟩+|111⟩)/√2.
- Each column is one simultaneous moment; gates in a column commute.

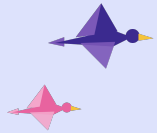
ENGINE

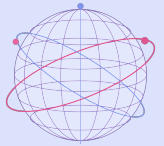
- Statevector for ideal runs, density matrix for noise; measurement collapse + `if_test` feed-forward.
- QASM parser reconstructs moments with per-qubit scheduling.

EVIDENCE

- Bell/GHZ, Deutsch–Jozsa, Bernstein–Vazirani, Grover, QFT, QPE, teleportation, noisy Bell.
- Web QFT matches QFTGate to 1e-12; Bell payload returns [0.5, 0, 0, 0.5].

Implementation & Tech Stack





FRONTEND

- React 19 + Vite + TypeScript; custom SVG circuit board with keyboard placement and ARIA labels.
- Hash routing keeps GitHub Pages deep links working without rewrites.

QUANTUM CORE

- TypeScript complex arithmetic; statevector and density simulation; QASM import/export.
- 13 templates; optimizer; static analyzer; Qiskit/Cirq/PennyLane exporters.

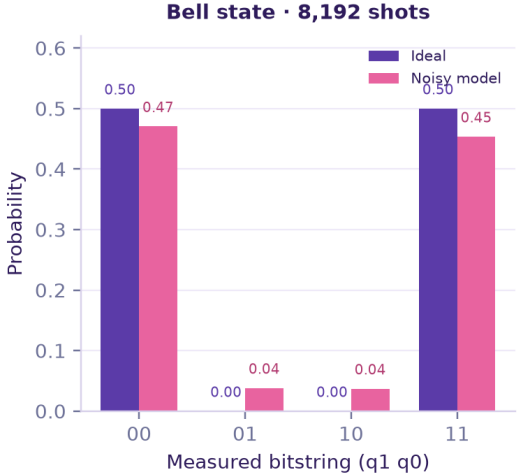
BACKEND & INFRA

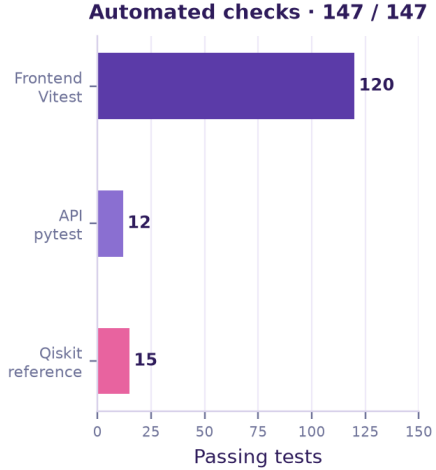
- FastAPI proxy keeps NVIDIA_API_KEY server-side; deterministic fallback when offline.
- Single-origin Docker service on Render; GitHub Actions deploys the static demo.
- 176-topic generated curriculum; BKT mastery and CSV/JSON educator export.

Results & Analysis

Real simulator output, not a mock-up

Qiskit 2.5.2 · Aer 0.18.0





WHAT THE DATA SHOWS

- Ideal Bell: 4,095 / 4,097 counts for |00⟩ / |11⟩ in 8,192 shots.
- Noise moves 7.7% of shots into |01⟩/|10⟩ and lowers purity; the UI surfaces entropy and concurrence.

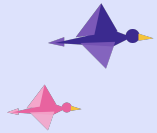
VERIFICATION

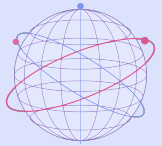
- 120 Vitest + 12 API + 15 Qiskit = 147/147 passing.
- QFT vs QFTGate: max error 1.2e-12; TypeScript strict; production build clean.

LIVE

- API health: numpy-statevector + NVIDIA; tutor verified end-to-end on Pages.
- Bell API returns [0.5, 0, 0, 0.5].

Impact & Real-World Applications





FOR LEARNERS

- Zero install, low bandwidth; first simulated circuit in under a minute.
- Immediate, grounded feedback when a gate is misplaced or a state is separable.

FOR EDUCATORS

- Class-level mastery map, pretest/posttest, normalised gain and challenge record.
- CSV/JSON export for spreadsheets or LMS import; no database required.

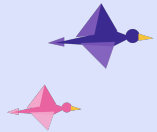
FOR INSTITUTIONS

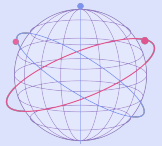
- Low-cost quantum lab on existing laptops; no GPU or cloud queue.
- Reproducible Qiskit artifacts bridge classroom work to real SDK workflows.

FOR INDUSTRY

- Faster onboarding for quantum software teams; a reusable simulator/analysis core.

Challenges & Future Scope





WHAT IS HARD

- Browser compute ceilings; density simulation is expensive beyond small registers.
- Approximate noise models; LLM latency and provider availability.
- Content authoring quality; multi-user persistence is not yet included.

HOW WE MITIGATE

- Statevector cap + density path; deterministic fallback; server-side key; localStorage.
- Content generator + agent review; explicit limitations in the README.

NEXT

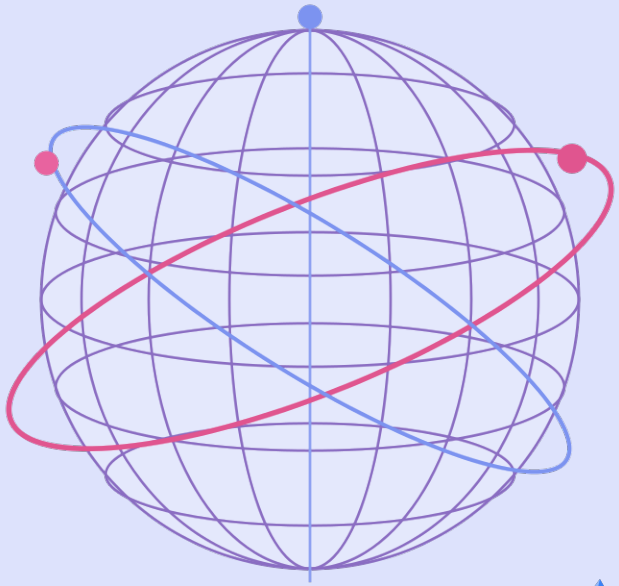
- Qiskit Runtime/hardware labs, adaptive BKT + spaced repetition, LMS/LTI integration.
- WebGPU simulator, VQE/QAOA/Shor modules, multilingual Chinnu, classroom pilots.

CLOSING

Thank You

Questions?

Inspiring Minds. Shaping Tomorrow.



GITHUB REPOSITORY
vtu-qiskit-fall-fest-2026/vtu-hc-q-matrix

LIVE DEMO / VIDEO
chinmayrerc.github.io/q-matrix

Live demo · https://chinmayrerc.github.io/q-matrix
API · https://q-matrix.onrender.com
Team code · VTU-QFF-HC-0024